CORE PURE / VECTORS, MATRICES & LINEAR TRANSFORMATIONS

Linear Transformations

Mark scheme 11 questions · 86 marks

Questions

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	Total	**86**	

Reading the mark scheme

These rules apply throughout, unless a note says otherwise.

Marks

Every mark is worth one, so **M1 A1** on a row is two marks. Only marks coded ft follow through.

Code	Meaning
M1	Method: a correct method, applied to this question. A slip doesn't cost it if the method is clear. A formula must be used, not just quoted.
dM1	Dependent method: as M1, but only if the method mark before it was earned, unless its note names another.
A1	Accuracy: a correct answer or step. It needs the method mark it follows.
A1ft	As A1, following through from an earlier answer that the note names.
A1 AG	The last mark of a part whose answer is printed in the question: complete working, with no errors.
B1	A correct value, fact or statement. It needs no method mark.
B1ft	As B1, following through from an earlier answer that the note names.
E1	A written explanation, interpretation or conclusion. Concluding a proof by induction needs every earlier mark.

Abbreviations

ft and AG qualify a code in the Marks column. The others appear in the notes.

ft	follow through	**AG**	answer given in the question
cao	correct answer only, with no follow-through	**cso**	correct solution only: no errors in the part
oe	or an equivalent form	**awrt**	any answer that rounds to this value
soi	seen, or implied by later working	**isw**	ignore working after a correct answer
SC	special case: credit outside the main method	**their**	the student's own earlier value
condone	a slip that doesn't cost the mark	[ ]	working the mark doesn't need
s.f.	significant figures	**d.p.**	decimal places

Reading the scheme

- Each row is a line of working, with the marks it earns beside it. Where a row carries two marks, each note starts with its code.
- **[*n*]** ends each part with its total.
- A shaded *Alternative* is a second method for the same total. Mark an answer by the method it follows.
- Any other valid method earns the equivalent marks, unless the question names a method or says *hence*.
- A sketch earns one B1 for each feature named below its figure.

Accuracy and answers

- Unless a question says otherwise, give answers to 3 significant figures, and angles in degrees to 1 decimal place. Angles in radians take 3 significant figures.
- A final answer with more figures is accepted if it rounds to the scheme's value; one with fewer is not. An answer spoiled by rounding too early loses its accuracy mark.
- Where a question asks for an exact answer, a decimal loses the final accuracy mark. Otherwise either is accepted.
- Wrong or missing units cost nothing unless the question asks for them.
- An omitted $\mathrm{d}\theta$, $\mathrm{d}x$ or $\mathrm{d}u$ in an integral is not penalised where the variable of integration is clear from the integrand and the limits. It is penalised where it is genuinely ambiguous: a substitution part-way through, two variables in play, or limits in the wrong variable.
- A correct answer with no working earns full marks, unless the question needs the working: show that, prove, hence, a method it names, or an exact answer. A wrong answer with no working earns nothing. Calculator answers are treated the same way.
- A correct answer, once reached, isn't spoiled by later working (isw).

Value of g

- Take $g = 9.8\,\mathrm{m\,s^{-2}}$ unless a question states a value, which is then the only one accepted. Answers that use $g = 9.8$ may be given to 2 or 3 significant figures.
- Where a question doesn't state g, an answer worked with $g = 10$ throughout earns the same marks, keeping any value printed in the question. The worked solutions list the final answers with $g = 10$.

Question	Scheme	Marks	Notes
1(a)	$\mathbf{P} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$	**B1**	
		[1]	
1(b)	$\mathbf{Q} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$	**B1**	
		[1]	
1(c)	$\mathbf{R} = \mathbf{QP}$	**B1**	B0 for $\mathbf{PQ}$.
		[1]	
1(d)	$\mathbf{R} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$	**B1ft**	ft their $\mathbf{P}$ and $\mathbf{Q}$, multiplied as $\mathbf{QP}$.
		[1]	
1(e)	Rotation about the origin	**B1**	Accept about O or $(0, 0)$.
	through 90° anticlockwise	**B1**	Accept $\frac{\pi}{2}$ anticlockwise or 270° clockwise. Direction required.
		[2]	
	Total for Question 1	**6**	

Question	Scheme	Marks	Notes
2(a)	$\begin{pmatrix}0 & -1\\ 1 & 0\end{pmatrix}\begin{pmatrix}-2 & 1 & 1\\ 1 & 1 & 5\end{pmatrix} = \dots$	**M1**	Applies **P** to the vertices, by multiplying or by the rule $(x, y) \mapsto (-y, x)$. One correct image is enough.
	$A'(-1,-2)$, $B'(-1,1)$, $C'(-5,1)$	**A1**	All three, as coordinates or column vectors.
		[2]	
2(b)	Rotation about the origin	**B1**	Needs rotation and the centre O or $(0,0)$.
	through 90° anticlockwise	**B1**	Accept $\frac{\pi}{2}$ anticlockwise or 270° clockwise. Condone +90° with no direction.
		[2]	
2(c)	$\mathbf{QR} = \begin{pmatrix}1 & 4\\ 3 & 0\end{pmatrix}$	**B1**	
		[1]	
2(d)	$\det(\mathbf{QR}) = 1 \times 0 - 4 \times 3$	**M1**	Uses $ad - bc$ on their 2×2 **QR**, or multiplies $\det \mathbf{Q}$ by $\det \mathbf{R}$. Condone one sign slip.
	$= -12$	**A1**	
		[2]	
2(e)	Area of $T = \frac{1}{2} \times 3 \times 4 = 6$	**B1**	
	Area of $T'' = \lvert\text{"}-12\text{"}\rvert \times 6$	**M1**	Multiplies their area of T by the modulus of their determinant from (d). Not for an area found from the vertices of T''.
	$= 72$	**A1ft**	ft their area of T and their determinant.
		[3]	
	Total for Question 2	**10**	

Question	Scheme	Marks	Notes
3(i)(a)	$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$	**B1**	
		[1]	
3(i)(b)	$\begin{pmatrix} \frac{1}{2} & \frac{\sqrt{3}}{2} \\ -\frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix}$	**B1**	Exact values, or awrt 0.866 for $\frac{\sqrt{3}}{2}$.
		[1]	
3(i)(c)	$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} \frac{1}{2} & \frac{\sqrt{3}}{2} \\ -\frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix} = \begin{pmatrix} -\frac{\sqrt{3}}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{\sqrt{3}}{2} \end{pmatrix}$	**B1ft**	ft their (a) times their (b), in that order. Not for the product in the reverse order.
		[1]	
3(ii)	$\det \begin{pmatrix} 2 & 5 \\ 1 & -1 \end{pmatrix} = -2 - 5 = -7$	**B1**	
	Area of $T = \dfrac{84}{\text{"}7\text{"}} = 12$	**M1 A1**	**M1** Divides 84 by the modulus of their determinant, or equates their scale factor times the area of T to 84.
	$\frac{1}{2}\left\lvert \det \begin{pmatrix} 4 & k-1 \\ 3 & -3 \end{pmatrix} \right\rvert$ $= \frac{1}{2}\lvert -12 - 3(k-1) \rvert$	**M1**	Finds the area of T in terms of k by any method, such as half a determinant of two side vectors or the shoelace formula. Condone sign slips.
	$= \frac{3}{2}(k+3)$	**A1**	$\frac{3}{2}(k+3)$ or $\frac{3}{2}\lvert k+3 \rvert$ oe.
	$\frac{3}{2}(k+3) = 12 \Rightarrow k = 5$	**A1**	$k = 5$ only; A0 if $k = -11$ is also given.
		[6]	
	Total for Question 3	**9**	

Question	Scheme	Marks	Notes
4(a)	$\begin{pmatrix}3\\1\end{pmatrix}, \begin{pmatrix}2\\3\end{pmatrix}$ Area of $S = \left\lvert\det\begin{pmatrix}3 & 2\\1 & 3\end{pmatrix}\right\rvert = \ldots$	**M1**	Finds the area of S by any valid method, such as the determinant of two adjacent side vectors from one vertex. Condone a sign slip.
	Area of $S = 7$	**A1**	
	"7" $\lvert\det \mathbf{P}\rvert = 35$	**M1**	Equates their area of S times $\det \mathbf{P}$, with or without the modulus, to 35, oe.
	$\det \mathbf{P} = p^2 + 2p - 3 = \pm 5$	**A1**	Correct $\det \mathbf{P}$ in any form, such as $(p+1)^2 - 4$, equated to 5 or to ± 5.
	$p^2 + 2p - 8 = 0 \Rightarrow p = \ldots$	**dM1**	Solves their quadratic in p by factorising, the formula, completing the square or a calculator.
	$p = 2$ ($p = -4$ rejected)	**A1**	$p = 2$ only, with -4 rejected. The case $\det \mathbf{P} = -5$ need not be solved.
		[6]	
4(b)	$\begin{pmatrix}0 & 1\\1 & 0\end{pmatrix}$	**B1**	
	$\begin{pmatrix}2 & 0\\0 & 1\end{pmatrix}$	**B1**	
	$\mathbf{Q} = \begin{pmatrix}2 & 0\\0 & 1\end{pmatrix}\begin{pmatrix}0 & 1\\1 & 0\end{pmatrix}$ $= \begin{pmatrix}0 & 2\\1 & 0\end{pmatrix}$	**B1ft**	ft their two matrices, stretch times reflection in that order, multiplied correctly.
		[3]	
4(c)	$\mathbf{R} = \mathbf{QP} = \begin{pmatrix}0 & 2\\1 & 0\end{pmatrix}\begin{pmatrix}3 & 2\\2 & 3\end{pmatrix}$ $= \begin{pmatrix}4 & 6\\3 & 2\end{pmatrix}$	**B1ft**	ft their $\mathbf{Q}$ and their p, with $\mathbf{QP}$, in that order, multiplied correctly.
		[1]	
	Total for Question 4	**10**	

Question	Scheme	Marks	Notes
5(i)	$\mathbf{A}^2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$	**B1**	
		[1]	
5(ii)	A reflection applied twice returns every point to itself, so $\mathbf{A}^2 = \mathbf{I}$ is consistent	**E1**	States that reflecting twice leaves every point unchanged, or that a reflection is self-inverse, and links this to $\mathbf{A}^2 = \mathbf{I}$.
		[1]	
5(iii)	$-\frac{1}{2}x + \frac{\sqrt{3}}{2}y = x$ $\frac{\sqrt{3}}{2}x + \frac{1}{2}y = y$	**M1**	Sets **A** times a general point (x, y) equal to (x, y), forming at least one equation. Condone slips in multiplying.
	$y = \sqrt{3}\,x$	**A1**	oe, such as $\sqrt{3}\,x - y = 0$.
		[2]	
5(iv)	Rotation about the origin	**B1**	Needs the centre, the origin or $(0, 0)$.
	through 120° anticlockwise	**B1**	Accept $\frac{2\pi}{3}$, or 240° clockwise. The direction is required.
		[2]	
5(v)	$\mathbf{BA} = \begin{pmatrix} -\frac{1}{2} & -\frac{\sqrt{3}}{2} \\ -\frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix}$	**B1**	The product in the order **BA**; **AB** scores B0.
		[1]	
5(vi)	$-\frac{1}{2}x - \frac{\sqrt{3}}{2}y = x$ $-\frac{\sqrt{3}}{2}x + \frac{1}{2}y = y$	**M1**	Sets their 2×2 matrix from (v) times (x, y) equal to (x, y), forming at least one equation. Condone slips in multiplying.
	$y = -\sqrt{3}\,x$	**A1ft**	oe. ft their matrix from (v), where its invariant points form a line.
		[2]	
	Total for Question 5	**9**	

Question	Scheme	Marks	Notes
6(a)	Stretch parallel to the y-axis, scale factor k	**B1**	Names a stretch; its direction and scale factor are not needed, but B0 if a wrong one is given.
	then reflection in the line $y = x$	**B1**	The reflection, after the stretch; its mirror line is not needed, but B0 if wrong. Reflection first scores B1 B1 only with the stretch parallel to the x-axis; otherwise B1 B0.
		[2]	
6(b)	$\mathbf{M} = \begin{pmatrix} 0 & k \\ 1 & 0 \end{pmatrix}$, $\det \mathbf{M} = -k$ Area $= k$	**B1**	Must be positive.
	$\mathbf{M}^{-1} = -\frac{1}{k}\begin{pmatrix} 0 & -k \\ -1 & 0 \end{pmatrix}$	**M1**	Attempts the inverse of their $\mathbf{M}$, by the determinant and adjugate, or as the inverses of the two matrices in reverse order. Condone one sign slip.
	$= \begin{pmatrix} 0 & 1 \\ \frac{1}{k} & 0 \end{pmatrix}$	**A1**	
		[3]	
6(c)	$\begin{pmatrix} 0 & k \\ 1 & 0 \end{pmatrix}\begin{pmatrix} x \\ \frac{x}{\sqrt{k}} \end{pmatrix} = \ldots$	**M1**	Applies their $\mathbf{M}$ to a general point of the line, such as $\left(x, \frac{x}{\sqrt{k}}\right)$ or (x, mx), or to its direction vector.
	$= \begin{pmatrix} \sqrt{k}\,x \\ x \end{pmatrix}$	**A1**	oe, such as $\left(\frac{kx}{\sqrt{k}}, x\right)$.
	$X = \sqrt{k}\,x,\ Y = x \Rightarrow Y = \frac{X}{\sqrt{k}}$ so the line is invariant	**A1 AG**	cso. Image shown to satisfy $y = \frac{x}{\sqrt{k}}$, or to be a multiple of the direction vector, with a conclusion. Via (x, mx), needs $m^2 = \frac{1}{k}$ and $m = \frac{1}{\sqrt{k}}$.
		[3]	
	Total for Question 6	**8**	

Question	Scheme	Marks	Notes
7	Compares with rotation about the y-axis: $\cos\theta = \frac{\sqrt{3}}{2}$, $\sin\theta = \ldots$	**M1**	Uses the y-axis rotation matrix, with $\sin\theta$ top right and $-\sin\theta$ bottom left, to find the sign of $\sin\theta$.
	$\sin\theta = -\frac{1}{2}$: Bilal ignored the sign $\theta = -30°$ (or $330°$)	**A1**	Error named as the sign of $\sin\theta$, and $\theta = -30°$ or $330°$ oe, or $30°$ clockwise viewed from the positive y-axis.
	Total for Question 7	**2**	

Question	Scheme	Marks	Notes
8(a)	$\mathbf{A}^5 = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$	**B1**	
		[1]	
8(b)	Rotation about the line $x = y = z$	**B1**	Rotation, with axis $x = y = z$ oe, such as the direction $\mathbf{i} + \mathbf{j} + \mathbf{k}$.
	through 120°, clockwise viewed from $(1, 1, 1)$ towards the origin	**B1**	Angle with a correct sense, oe such as anticlockwise viewed from the origin towards $(1, 1, 1)$, or 240° anticlockwise from $(1, 1, 1)$. B0 without a sense.
		[2]	
8(c)	$\mathbf{B} = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$	**B1**	
		[1]	
8(d)	Q' is $(-1, 4, 2)$	**B1ft**	ft their $\mathbf{B}$ applied to $(4, -1, 2)$. Condone a column vector.
		[1]	
	Total for Question 8	**5**	

Question	Scheme	Marks	Notes
9(a)	$\alpha = 225$	**B1**	Or any $225 + 360n$, such as -135.
		[1]	
9(b)	$225k = 360n \Rightarrow 5k = 8n \Rightarrow k = \ldots$	**M1**	Sets k times their α equal to a multiple of 360, or to 360 times an integer, and attempts the least k.
	$k = 8$	**A1**	
		[2]	
9(c)	$\mathbf{N} = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$	**B1**	
		[1]	
9(d)	$\mathbf{NM}\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} = \mathbf{N}\begin{pmatrix} -2\sqrt{2} \\ 2 \\ -\sqrt{2} \end{pmatrix} = \ldots$	**M1**	Multiplies $(1, 2, 3)$ by $\mathbf{M}$, then by their reflection matrix $\mathbf{N}$, or by their $\mathbf{NM}$, in that order. Condone slips.
	$= \begin{pmatrix} 2\sqrt{2} \\ 2 \\ -\sqrt{2} \end{pmatrix} \Rightarrow B = (2\sqrt{2}, 2, -\sqrt{2})$	**A1 AG**	Correct products with $\mathbf{M}$ and the correct $\mathbf{N}$, surds simplified, leading without error to the printed coordinates.
		[2]	
9(e)	$\lvert\overrightarrow{OA}\rvert = \sqrt{14}$, $\lvert\overrightarrow{OB}\rvert = \sqrt{8+4+2} = \sqrt{14}$	**B1**	
	$\cos\angle AOB = \frac{2\sqrt{2}+4-3\sqrt{2}}{\sqrt{14}\sqrt{14}}$	**M1**	Divides a three-term scalar product of $\overrightarrow{OA}$ and $\overrightarrow{OB}$ by their magnitudes, or uses the cosine rule with their $\lvert AB\rvert$. Condone slips.
	$= \frac{4-\sqrt{2}}{14} \Rightarrow \angle AOB = 79.4°$	**A1 AG**	Correct $\cos\angle AOB = \frac{4-\sqrt{2}}{14}$ oe, leading without error to $79.4°$.
		[3]	
9(f)	Area $= \frac{1}{2}\sqrt{14}\sqrt{14}\sin 79.4°$	**M1**	Uses $\frac{1}{2}\lvert\overrightarrow{OA}\rvert\lvert\overrightarrow{OB}\rvert\sin\angle AOB$ with their magnitudes and $79.4°$, a more accurate angle or the exact sine.
	$= 6.88$	**A1**	
		[2]	
	Total for Question 9	**11**	

Question	Scheme	Marks	Notes
10(a)	$\mathbf{R} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{pmatrix}$ $\Rightarrow \mathbf{R}^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & \sin\theta \\ 0 & -\sin\theta & \cos\theta \end{pmatrix}$	**B1**	Correct $\mathbf{R}$ or $\mathbf{R}^{-1}$, stated or used.
	$\mathbf{A} = \mathbf{R}^{-1}\mathbf{B} = \ldots$	**M1**	Uses $\mathbf{A} = \mathbf{R}^{-1}\mathbf{B}$, not $\mathbf{B}\mathbf{R}^{-1}$, with their $\mathbf{R}^{-1}$ of a rotation matrix's form, and attempts the product. Condone slips in multiplying.
	$\mathbf{A} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & \cos^2\theta + \sin^2\theta \\ 0 & \sin^2\theta + \cos^2\theta & 0 \end{pmatrix}$ $= \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$, so no θ	**A1 AG**	Correct product simplified by $\sin^2\theta + \cos^2\theta = 1$ to a matrix free of θ, with a conclusion that $\mathbf{A}$ is independent of θ.
		[3]	
10(b)	Reflection in the plane $y = z$	**B1**	Needs reflection and the plane, $y = z$ oe.
		[1]	
	Total for Question 10	**4**	

Question	Scheme	Marks	Notes
11(a)	Rotation about the origin	**B1**	
	through angle θ anticlockwise	**B1**	Accept clockwise through $2\pi - \theta$.
		[2]	
11(b)(i)	$\det(\mathbf{M}) = \cos^2\theta + \sin^2\theta = 1$	**B1**	
		[1]	
11(b)(ii)	$\lvert\det(\mathbf{M})\rvert = 1 \Rightarrow$ area is preserved	**E1**	States that area is preserved, or the area scale factor is 1, linked to the determinant.
	$\det(\mathbf{M}) > 0 \Rightarrow$ orientation is preserved	**E1**	States that orientation is preserved, or that T involves no reflection, linked to the determinant being positive.
		[2]	
11(c)	$n = 1$: $\cos 1\theta = \cos\theta$, $\sin 1\theta = \sin\theta$ $\Rightarrow \mathbf{M}^1 = \mathbf{M}$, so true for $n = 1$	**B1**	Formula evaluated at $n = 1$ and seen to give $\mathbf{M}$.
	Assumes true for $n = k$: $\mathbf{M}^{k+1} = \mathbf{M}^k\mathbf{M}$	**M1**	Assumes the result for $n = k$ and writes $\mathbf{M}^{k+1}$ as $\mathbf{M}^k\mathbf{M}$ or $\mathbf{M}\mathbf{M}^k$. Not if the result for $n = k$ is stated as fact.
	$\begin{pmatrix}\cos k\theta & -\sin k\theta \\ \sin k\theta & \cos k\theta\end{pmatrix}\begin{pmatrix}\cos\theta & -\sin\theta \\ \sin\theta & \cos\theta\end{pmatrix} = \ldots$	**M1**	Multiplies the assumed form of $\mathbf{M}^k$ and $\mathbf{M}$, in either order, each entry a sum of two products. Condone sign slips.
	$\begin{pmatrix}c & -s \\ s & c\end{pmatrix}$, where $c = \cos k\theta\cos\theta - \sin k\theta\sin\theta$ $s = \sin k\theta\cos\theta + \cos k\theta\sin\theta$	**A1**	All four entries correct, unsimplified.
	$\Rightarrow \mathbf{M}^{k+1} = \begin{pmatrix}\cos(k+1)\theta & -\sin(k+1)\theta \\ \sin(k+1)\theta & \cos(k+1)\theta\end{pmatrix}$	**A1**	cso. All four entries in the form for $n = k + 1$, using the addition formulae.
	True for $n = 1$; true for $n = k$ $\Rightarrow$ true for $n = k + 1$ $\Rightarrow$ true for all positive integers n	**E1**	Concludes with all three points, true for $n = 1$, true for $n = k$ implies true for $n = k + 1$, and so true for all positive integers n.
		[6]	

Question 11 continued

Question	Scheme	Marks	Notes
11(d)	Rotation through $n\theta$ anticlockwise about the origin	**B1ft**	ft their description in (a) with θ replaced by $n\theta$. Needs the centre, angle and sense. Condone the angle not reduced modulo 2π.
		[1]	
	Total for Question 11	**12**	